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Microaerobic control of biogas sulphide content during sewage sludge digestion by using biogas production and hydrogen sulphide concentration

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HIGHLIGHTS

- Microaerobic conditions are applied to controlling the biogas sulphide content.
- Biogas production is used to regulate the oxygen supply.
- Biogas sulphide content is used to optimise the oxygen supply by a PID controller.
- Biogas production can be an efficient regulating parameter at steady sulphur load.
- Biogas sulphide content can be a precise regulating parameter in all conditions.

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ABSTRACT

This paper presents the potentials of using biogas production and hydrogen sulphide concentration as the parameters to regulate the oxygen supply to microaerobic reactors in order to control the biogas sulphide content. Research was carried out in two identical bioreactors of 200 L at 35 °C and 19 d of hydraulic retention time. The feed consisted of mixed sludge from a municipal wastewater treatment plant with variable organic and sulphur load. The oxygen flow rate was automatically adjusted according to the biogas sulphide content (which ranged from 0.62 to 0.24%v/v) by a feedback Proportional-Integral-Derivative controller. The target hydrogen sulphide concentration (0.01%v/v) was achieved in 4.0–5.5 h. The micro-oxygenation level reached was considered to be the optimum in the short-medium term, since it kept the removal efficiency above 99% and minimised the oxygen concentration in the biogas during the days following the controller application. Specifically, the average biogas oxygen content was 0.09%v/v. Subsequently, biogas production was used as the parameter to regulate the oxygen supply. When the biogas sulphide content was around 0.33 and 0.50%v/v, approximately 3.5 and 5.0 NL of oxygen were supplied per N m^3 of biogas (respectively). An average sulphide removal efficiency of 99%, and oxygen concentrations in the biogas of less than 0.08%v/v were achieved. Biogas production could be employed to develop precise control strategies during microaerobic digestion under variable organic load and steady sulphur load. Under unstable sulphur load, biogas sulphide content should be used instead.

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1. Introduction

Biogas is a versatile and renewable energy source produced mainly by the anaerobic digestion of sewage sludge [1]. Methane and carbon dioxide are the main constituents, but it also contains significant quantities of undesirable compounds such as hydrogen sulphide, whose concentration can reach 1.0%v/v [2]. It is produced by sulphate-reducing and acidogenic bacteria mainly from sulphate and proteins, respectively [3]. Hydrogen sulphide escapes with the biogas, and has detrimental impacts on society and health, environment, and installations for biogas utilisation. Namely, it causes bad odours and eye damage when values are below 0.01%v/v, and even death when above 0.03%v/v [4]. Manufacturers of combined heat and power (CHP) production units recommend limiting values to between 0.01 and 0.03%v/v in order to prevent corrosion in piping systems and equipment. However, short peaks can occasionally be accepted [5]. Therefore, biogas

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sulphide content has to be controlled in order to prevent damage and fulfil the quality standards required in consistence with the chosen application of the biogas.

Recently, there has been wide interest in desulphurisation biotechnologies, since they are an effective and environmentally friendly way of solving the problems of large investments and operational costs incurred by physicochemical processes [6]. Inside them, hydrogen sulphide is removed by sulphide-oxidising bacteria (SOB), which obtain energy by employing sulphide as the electron donor and oxygen as the electron acceptor [7]. The pathway of sulphide biological oxidation inside bioreactors has been suggested as: $H_2S \rightarrow S^0 \rightarrow S_2O_3^{2-} \rightarrow S_4O_6^{2-} \rightarrow S_3O_6^{2-} \rightarrow SO_3^{2-} \rightarrow SO_4^{2-}$ [8]. Sulphide can be also chemically oxidised [9]. Among the bioprocesses, many investigators have turned to microaerobic removal, which consists of supplying limited amounts of oxygen (or air) directly into the anaerobic bioreactor. This is possible because SOB are present in numerous substrates treated by anaerobic digestion [2]. Thus, no additional unit (such as a bioscrubber, biofilter or biotrickling filter) is needed. It is important to note that these microaerobic reactors yield similar to the anaerobic ones [10], or even better [11].

During digestion under microaerobic conditions, SOB colonise the headspace of the reactor and oxidise hydrogen sulphide by using the oxygen that reaches this area, independently of both the oxygen dosing point and mixing method [12]. Díaz et al. [12] demonstrated that the most efficient reactor configuration in order to microaerobically desulphurise biogas involves injecting the oxidant agent into the headspace and implementing liquid recirculation as the mixing method. Thus, the oxygen consumption in undesired processes was minimised. As a result, elemental sulphur accumulates all over the gas space [13]; it must be taken into account that both reactants (air or oxygen) are supplied in limited amounts in order to minimise both their concentration in the biogas, and the operating costs. With regard to this, it is worth noting that the mixtures of methane–oxygen formed inside microaerobic reactors are far from being explosive, since the limits of methane flammability in air are 5.0–15.0%v/v. Additionally, it should be considered that gases such as carbon dioxide reduce this concentration range [4].

The digester response to the presence of limited amounts of oxygen in terms of biogas composition and, more specifically, in terms of hydrogen sulphide and oxygen concentration, has been proved to be really rapid [14]. Moreover, Jenicek et al. [15] and Díaz et al. [16] highlighted the steady dynamic behaviour of microaerobic reactors treating solid wastes and sewage sludge, respectively, within a wide range of hydrogen sulphide concentrations. They also pointed out that microaeration decreases the heating value of the biogas (that is, methane concentration) due to the presence of nitrogen. Accordingly, though it is expensive, the most profitable oxidant agent is pure oxygen. Nonetheless, it must be taken into account that not all the oxygen supplied to a digester is often used therein [16,17]. With regard to this, it is worth noting that unless the biogas is used for CHP or boilers, the presence of oxygen should be avoided, since it is expensive to remove [18]. In Europe, if biogas is to be used as a vehicle fuel or injected into fuel cells or natural gas networks, its oxygen content must not exceed concentrations of 1.0 and 3.0%v/v, respectively [5].

On a full-scale, since hydrogen sulphide production can vary according to the feeding composition, the micro-oxygenation rate must be periodically regulated in order to avoid a lack or surplus of oxygen in the biogas, while maintaining the quality standards required. For this purpose, the key issue is to find a variable capable of providing a precise control of the oxygen supply; little research efforts have been made within this context. Oxidation–reduction potential (ORP) has been reported as an accurate regulation parameter of oxygen dosing in order to eliminate sulphide toxicity [19], maximise sulphur recovery [20], and even desulphurise biogas [21]. However, its response to micro-oxygenation can be insufficient to develop a reliable control of the hydrogen sulphide concentration in biogas during digestion [16]. On the basis that the gaseous sulphide and biogas production both increase and decrease concurrently with the organic loading rate (OLR) due to rises and decreases in fermentative activity, Díaz et al. [22] proposed to regulate the oxygen flow rate according to the biogas production. They found a linear correlation between the ratio of the oxygen supply to the biogas flow rate, and the biogas sulphide content. Under steady hydrogen sulphide concentration, biogas production would be used to develop a reliable and consistent control strategy. Otherwise, the biogas sulphide content could be an efficient regulating parameter of the micro-oxygenation.

The performance of oxygen utilisation inside a bioreactor is expected to vary with time due to increasing elemental sulphur accumulation in the headspace. This could alter the oxygen transfer conditions and affect biological oxidation rates as a result of the change in the growing conditions. Therefore, a control approach using hydrogen sulphide concentration in biogas as the regulating parameter of the oxygen supply would automatically absorb changes in both performance of oxygen utilisation in the digester and biogas production. Proportional-Integral-Derivative (PID) control is the standard automatic controller in industrial settings. Among this type of control systems, the feedback PID controller determines an input variable to the control process by using the measurement of an output variable [23]. Besides being applicable to many real-world control problems, the PID controller is simple, intuitive, efficient, and reliable for processes with steady dynamic behaviour [24]. Therefore, it could be successfully applied to control the biogas sulphide content in microaerobic digesters by using biogas sulphide content as the regulating parameter of the oxygen flow rate.

The aim of this study is to investigate the feasibility of using biogas production and hydrogen sulphide concentration in biogas to regulate the oxygen flow rate, thereby achieving a consistent and efficient control of the hydrogen sulphide concentration during microaerobic digestion.

2. Materials and methods

2.1. Pilot-scale digesters

Two identical continuous stirred tank reactors called R1 and R2 with a working volume of 200 L and a headspace of 50 L were operated under mesophilic conditions and 19 ± 1 d of hydraulic retention time (Fig. 1). Temperature was maintained by an electric resistor. The pressure control was performed hydraulically; an electro-valve regulated the biogas outflow.

Mixed sludge with variable composition was transported weekly to the pilot plant from a municipal wastewater treatment plant. It was stored at 4 °C, and was fed continuously from two stirred tanks at room temperature into both digesters by peristaltic pumps. Sludge recirculation at a rate of 50 L/h ensured mixing. Pure oxygen from a cylinder was injected by means of two mass flow controllers (Bronkhorst EL-FLOW Select) into the headspace or the recirculation stream, depending on the operational stage (Fig. 1).

2.2. Monitoring and experimental analysis

Digestion pressure and temperature were monitored. Biogas was quantified by the displacement of a fixed liquid volume, and its composition was determined by a CP-4900 Micro-GC [12]. The detection limit (DL) for hydrogen sulphide was 0.001%v/v.

The feed and the digested sludge were sampled weekly for total and soluble chemical oxygen demand, total solids, volatile solids

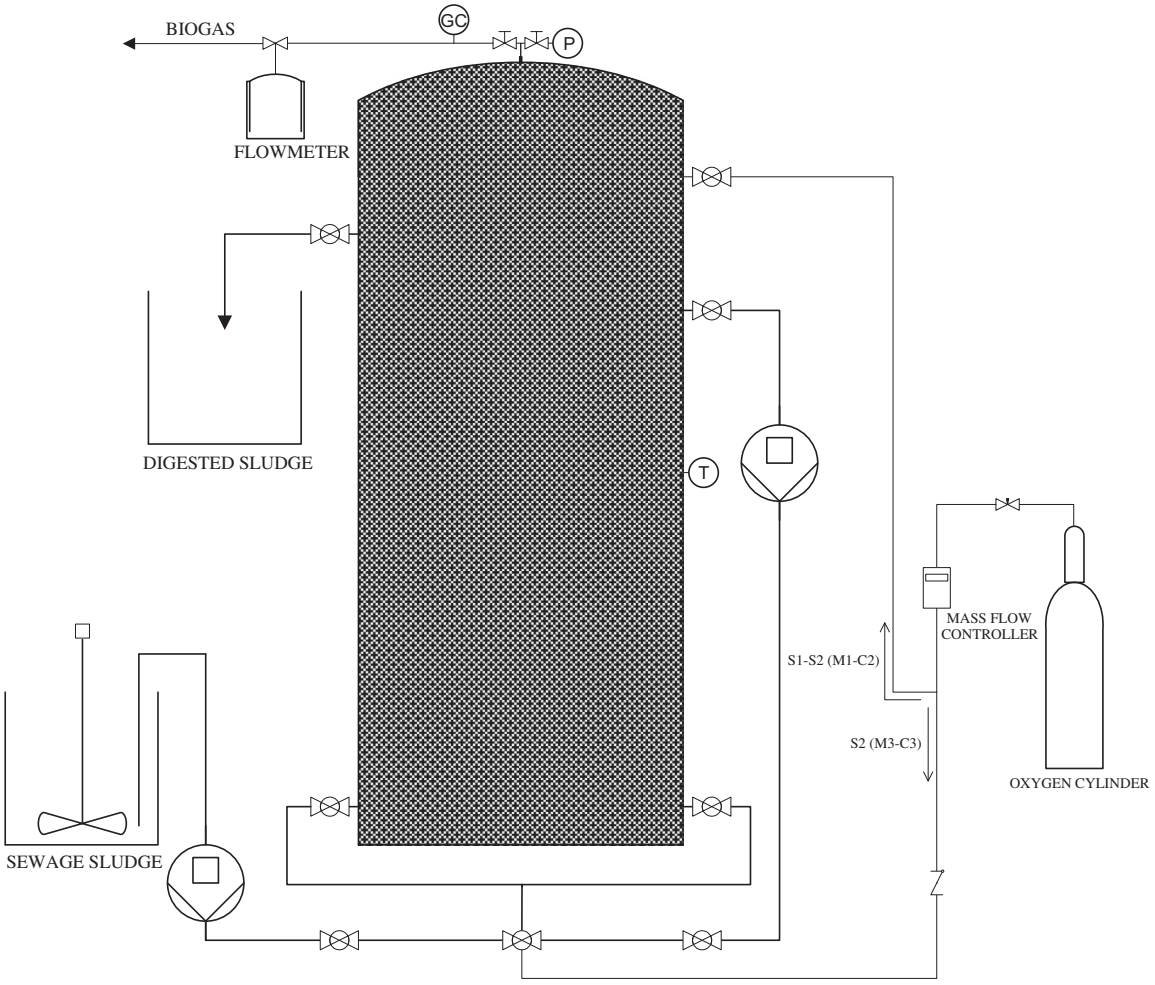


Fig. 1. R1 and R2 diagram.

(VS), volatile fatty acids, total Kjeldahl nitrogen, ammonia, sulphide, sulphate and thiosulphate analysis. This last anion was measured by high performance liquid chromatography, according to the procedure described by van der Zee et al. [9]; the rest of the parameters were determined according to standard methods [25].

2.3. Design of the feedback PID controller

The object of the feedback PID controller was to control the microaerobic process of biogas desulphurisation. Therefore, it was used to set the oxygen flow rate according to the error (e) between the measurement of hydrogen sulphide concentration and the target concentration of hydrogen sulphide according to the Eq. (1).[1] It was implemented using *macros* in *Microsoft Office Excel*.

$$\textit{Oxygen flow rate}(t) = \textit{Oxygen flow rate}(t-T) + g_0 e(t) + g_1 e(t-T) + g_2 e(t-2T) \quad (1)$$

The sampling time of the controller (T) was set at 30 min. It was fixed according to the well-known steady dynamic behaviour of the digester and the lax control of the biogas sulphide content required in this study. A lower value of T was considered unnecessary, whereas a larger value could be insufficient to achieve an efficient control of hydrogen sulphide concentration in the biogas, due to the controller acting too infrequently. The target biogas sulphide content or set-point was set at 0.01%v/v, which is the lowest concentration limit recommended in CHP [5].

The control parameters[2] (K_p, T_i, and T_d) were set at 1, 100 and 15 min, respectively. They were estimated manually based on the vast experience of the authors in operating the pilot plant, and the individual effects of the three actions of the PID controller on the performance thereof.[3] In order to minimise the surplus of oxygen, and to operate as close as possible to the linear range, K_p must be low, and T_i must be fairly high. Although the steady state error decreases with decreasing T_i, the process to be controlled in this study was not expected to present steady-state error (it could be eliminated). Therefore a low value of T_i was considered to be unnecessary; besides, it could have degraded the stability of the controller. On the other hand, due to high values of T_d improving both the stability and the drop time both, and taking into account that the reactor presented a steady dynamic behaviour, T_d was set relatively high.

Regarding the preceding paragraph, it should be mentioned that a wide range of tuning methods exist [27]; however, they were not applicable to this study due to the fact that the model of the process (*oxygen supply = f(biogas sulphide content)*) had not been estimated, and the relationship between the hydrogen sulphide concentration and oxygen supply was mathematically inverse

[1] See the appendix for more information about the scheme of the controller.

[2] See the appendix for more information about the mathematical description of the controller.

[3] See the appendix for a description of the effects of the three actions of the controller on its performance.

(which did not permit the application of tuning rules based on empirical procedures). As a result, the accuracy range of the linear relationship between the oxygen supply and the biogas sulphide content could not be estimated. However, this was considered to be unnecessary for the adjustment of the control parameters, since the achievement of the research objective was ensured due to the great experience of the authors in operating the digester. Moreover, it must be highlighted that the linearity premise was the basis for the estimation of the control parameters (see above).

2.4. Experimental procedure

The research was divided into two stages, S1 and S2. In S1, the oxygen flow rate was regulated according to the biogas sulphide content by implementing the PID controller presented above. In S2, biogas production was used as the parameter to regulate the oxygen supply. Both stages were in turn divided into several periods, designated as A (anaerobic periods), M (microaerobic periods in which the oxygen flow rate was maintained constant or regulated arbitrarily) and C (microaerobic periods in which the oxygen supply was regulated according to the biogas production or the hydrogen sulphide concentration).

2.4.1. Stage S1

The first stage of the study (S1) was carried out in R1 (Table 1). The oxygen supply was always injected into the headspace. As shown in Table 1, the controller was tested six times, over approximately 168 d, in order to evaluate its performance in the long term. Moreover, it was tested under different starting concentrations of hydrogen sulphide, hence the pauses introduced between the last three trials. Substantial changes in both the biogas sulphide content (which arose from the variability of the feeding) and the dynamic of the desulphurisation process were not expected in the short term. It is worth noting that the bioreactor was operated most of the time under microaerobic conditions during the pause periods. As a result, the amount of elemental sulphur attached to the headspace presumably increased.

Except for C1, the rest of the C periods (from C2 to C6) were started after a 2-day anaerobic period in which the hydrogen sulphide concentration remained stationary; namely, *oxygen flow rate*$(t - T) = 0$ and $e(t) \approx e(t - T) \approx e(t - 2T)$. Conversely, a concentration peak of hydrogen sulphide was induced at the beginning of C1 (that is, $e(t - 2T) < e(t - T) < e(t)$) by stopping the oxygen supply shortly before the start of the controller (namely, *oxygen flow rate*$(t - T) = 0$). The purpose of this was to evaluate its suitability in dealing with concentration peaks.

2.4.2. Stage S2

As shown in Table 2, S2 was divided into three trials, according to the different hydrogen sulphide concentrations in the biogas during the A periods, and the oxygen dosing point during the C and M periods. A total of eight operational periods were included. During the first two trials, R1 and R2 were operated in parallel and were subjected to the same variations in the operational conditions in terms of presence or absence of oxygen. During M1, C1 and C2, the oxygen was injected into the headspace (Fig. 1). Thus, only R1 was operated with oxygen supply to the recirculation stream (last trial).

3. Results and discussion

3.1. Stage S1: control based on biogas sulphide content

As illustrated in Fig. 2a, b, d, f, h and j, the controller started injecting oxygen at $t = 1$ h according to the last three measurements of hydrogen sulphide concentration recorded in the respective A periods (Table 1). Even in C1, when the controller was confronted with a sudden increase in hydrogen sulphide concentration, biogas sulphide content dropped below the set-point (0.01%v/v) in a time range from 4.0 to 5.5 h, subsequently stabilising at zero. Meantime, the biogas oxygen content rose from 0.02%v/v to, at most, 0.21%v/v (C5 in Table 1). This can be explained by the fact that minute amounts of oxygen entering the digester are inevitable under anaerobic conditions due to interactions with the surroundings as a result of sludge pumping. Oxygen content remained around 0.05%v/v during most of the C periods. Hence, the controller performance was successful, since the biogas was rapidly desulphurised and accurately adjusted (or optimised) the

Table 1
Sequence of variations applied to R1 during S1 and digester response. A: anaerobic period, M: microaerobic period in which the oxygen supply was maintained constant, C: microaerobic period in which the oxygen supply was regulated according to the hydrogen sulphide concentration, HS: headspace.

Trial	Period	Oxygen dosing point	Oxygen supply (NL/N m^3_{biogas})	VS (g/L)	Biogas production (N $m^3/m^3/d$)	Hydrogen sulphide (%v/v)[b]	Oxygen (%v/v)	Methane (%v/v)
1	A1[a]	–	0	19	0.75	0.17	0.02	65.3
	C1	HS	4.8–12.6	19	0.75	<DL in 4.5 h	0.04	65.6
2	A2[a]	–	0.0	16	0.60	0.58	0.02	67.1
	C2	HS	6.1–19.1	16	0.61	0.01 in 4.0 h	0.05	67.1
	M2	HS	12.6	15	0.58	<DL	0.05	65.9
3	A3[a]	–	0	21	0.77	0.58	0.02	61.8
	C3	HS	4.4–12.3	22	0.81	0.01 in 4.5 h	0.05	62.0
	M3	HS	7.9	22	0.81	<DL	0.05	64.2
4	A4[a]	–	0	22	0.84	0.52	0.02	62.5
	C4	HS	3.8–10.7	27	0.92	<DL in 4.5 h	0.05	62.2
	M4	HS	7.3	27	0.89	<DL	0.05	62.4
52-d pause		HS						
5	A5[a]	–	0	23	0.61	0.58	0.02	63.8
	C5	HS	4.2–15.9	23	0.59	<DL in 5.0 h	0.15	63.4
	M5	HS	12.1	23	0.60	<DL	0.19	63.5
69-d pause		HS						
6	A6[a]	–	0	35	1.04	0.26	0.02	61.2
	C6	HS	1.8–7.3	35	1.01	<DL in 5.5 h	0.03	61.3
	M6	HS	5.9	34	0.94	<DL	0.12	62.0

[a] Includes the three measurements for the calculation of the first oxygen supply (see Fig. 2a, b, d, f, h and j).
[b] DL = 0.001%v/v.

Table 2
Sequence of variations applied during S2 and response of the digesters. A: anaerobic period, M: microaerobic period in which the oxygen supply was adjusted arbitrarily, C: microaerobic period in which the oxygen supply was regulated according to the biogas production, HS: headspace, RS: recirculation stream.

Trial	Period	Reactor	Oxygen dosing point	Oxygen supply (NL/N m^3_{biogas})	VS (g/L)	Biogas production (N $m^3/m^3/d$)	Hydrogen sulphide (%v/v)[a]	Oxygen (%v/v)	Methane (%v/v)
1	A1	R1	–	0	35	1.03	0.25	0.01	62.9
		R2	–	0	36	1.01	0.37	0.01	61.1
	M1	R1	HS	5.7	35	0.95	<DL	0.11	62.2
		R2	HS	7.9	31	0.89	<DL	0.23	61.3
	C1	R1	HS	3.9	28	0.77	0.03	0.03	62.4
		R2	HS	3.9	26	0.72	0.01	0.04	61.9
2	A2	R1	–	0	28	0.75	0.54	0.01	62.5
		R2	–	0	22	0.57	0.47	0.01	62.0
	C2	R1	HS	5.0	31	0.88	<DL	0.02	63.1
		R2	HS	4.9	29	0.82	<DL	0.06	62.5
3	A3	R1	–	0	33	–	0.39	0.01	62.5
	M3	R1	RS	9.4	33	0.95	<DL	0.03	63.7
	C3	R1	RS	3.8	32	0.90	0.02	0.02	64.5

[a] DL = 0.001%v/v.

micro-oxygenation level, independently of the hydrogen sulphide flow rate (which varied widely, as shown in Table 1).

The hydrogen sulphide production was significantly higher in C3 than in C2, and the starting concentration was 0.58%v/v in both periods. Nonetheless, the optimum micro-oxygenation levels and the resulting biogas oxygen contents were almost equal (Table 1). Therefore, although the performance of oxygen utilisation inside the digester certainly changed, the controller continued providing successful results. Presumably, the longer biogas residence time increased the oxygen transfer to the liquid phase during C2. Thus, the oxygen consumption in unidentified processes rose, thereby reducing the amount available to SOB in the headspace, where the process occurred. Hence the oxygen demand was higher, and consequently a larger amount of oxygen was supplied by the controller. At this point, it is worth noting that neither the concentration of dissolved sulphides decreased, nor the sulphate or the thiosulphate content rose in S1 (data not shown). This was indeed as expected, according to Díaz et al. [12], as was the low oxygen transfer rate to the liquid phase. It should be considered that a limited contact area existed between the liquid media and the oxygen-rich biogas. Besides, the methane concentration was not negatively affected (Table 1).

The controller response (oxygen flow rate) outlined a similar profile in all the C periods (Fig. 2a, b, d, f, h and j); it corresponded to a perfectly predicted PID controller behaviour. Besides, both oxygen supply and hydrogen sulphide concentration (even biogas oxygen content) reached the stationary state simultaneously. As shown in Fig. 2a, b, d, h and j, once the set-point was achieved, the controller set oxygen flow rates which were negligibly different as a result of the minute variations in biogas sulphide content (namely, due to the steady dynamic behaviour of the reactor). Therefore, the micro-oxygenation level reached once the gaseous sulphide concentration decreased below the set-point was considered to be the optimum in order to maintain, at least in the short term, high removal efficiencies of hydrogen sulphide and low surpluses of oxygen; hence the shorter duration of C4 in relation to C1, C2, C3, C5 and C6. In C4, the controller was stopped once the set-point was achieved (Fig. 2f).

As noted, the oxygen supplies reached in C2, C3, C4, C5 and C6 just after achieving the set-point were considered to be the optimum. Therefore, they were maintained during M2, M3, M4, M5 and M6, respectively (Fig. 2c, e, g, i and k). It was deduced, from the variations in biogas production and oxygen concentration, that the hydrogen sulphide flow rate presumably fluctuated during M2, M5 and M6. However, the putative optimum oxygen flow rates effectively sufficed to maintain the removal efficiency over 99% after all the C periods (Table 1). Furthermore, they were also low enough to maintain biogas oxygen contents below 0.23%v/v. As a result, the biogas methane content was preserved. It is worth highlighting that M2 was prolonged for 19 d, while M3, M4, M5 and M6 lasted only 4–6 d. After the putative optimum oxygen supply was confirmed to be over 1 hydraulic retention time (M2), a 4-day M period was considered sufficient to conclude that successful performance could be maintained in the short-medium term.

Though unnecessary during this study, due to the successful results obtained from the application of the PID controller, the refinement of some or all the control parameters could become necessary in the longer term in order to maintain the efficiency of the control. For example, the oxygen demand could rise due to the increasing deposition of elemental sulphur in the headspace and the consequent reduction in oxygen transfer rate, thus requiring (for instance) a rise in K_p.

3.2. Stage S2: control based on biogas production

As illustrated in Fig. 3a and c, R1 and R2 operated under anaerobic conditions (period A1 in Table 2) until the 4th day, and the hydrogen sulphide concentration in the biogas was around 0.25 and 0.37%v/v, respectively. Thereafter, oxygen was injected into the headspace of both reactors (M1), and the flow rate was set according to the ratio of oxygen supplied to hydrogen sulphide produced, as adjusted in the last trial of S1, which was approximately 2.0 (v/v). In R2, the micro-oxygenation level was reduced stepwise during M1 in order to minimise further the oxygen concentration in the biogas, while maintaining the biogas sulphide content below 0.01%v/v. It was achieved on the 10th day by injecting 3.5 NL of oxygen per N m^3 of biogas. As a result, the oxygen supply to both reactors during C1 was readjusted daily (or occasionally every two or three days) based on the biogas flow rate measured over the previous 24 h and according to that relationship.

R2 apparently utilised the oxygen more efficiently; as noted, the hydrogen sulphide concentration in R2 during A1 was significantly higher than in R1. Nevertheless, since both digesters were fed with the same sludge and operated under almost the same conditions (slight differences occurred due to the significant variability in feeding), very similar biogas compositions were generally recorded. In fact, considering the profile of both the concentration and the flow rate of oxygen in R2 during M1, as well as the subsequent results (presented below), the hydrogen sulphide concentration in R2 presumably approached that in R1 after the 4th day (Fig. 3c). Unfortunately, sulphur-containing anions were not determined during M1. That presumable delay in the change in the hydrogen sulphide concentration was related to the variability of the sewage sludge, which is reflected in Table 2, Fig. 3b and d.

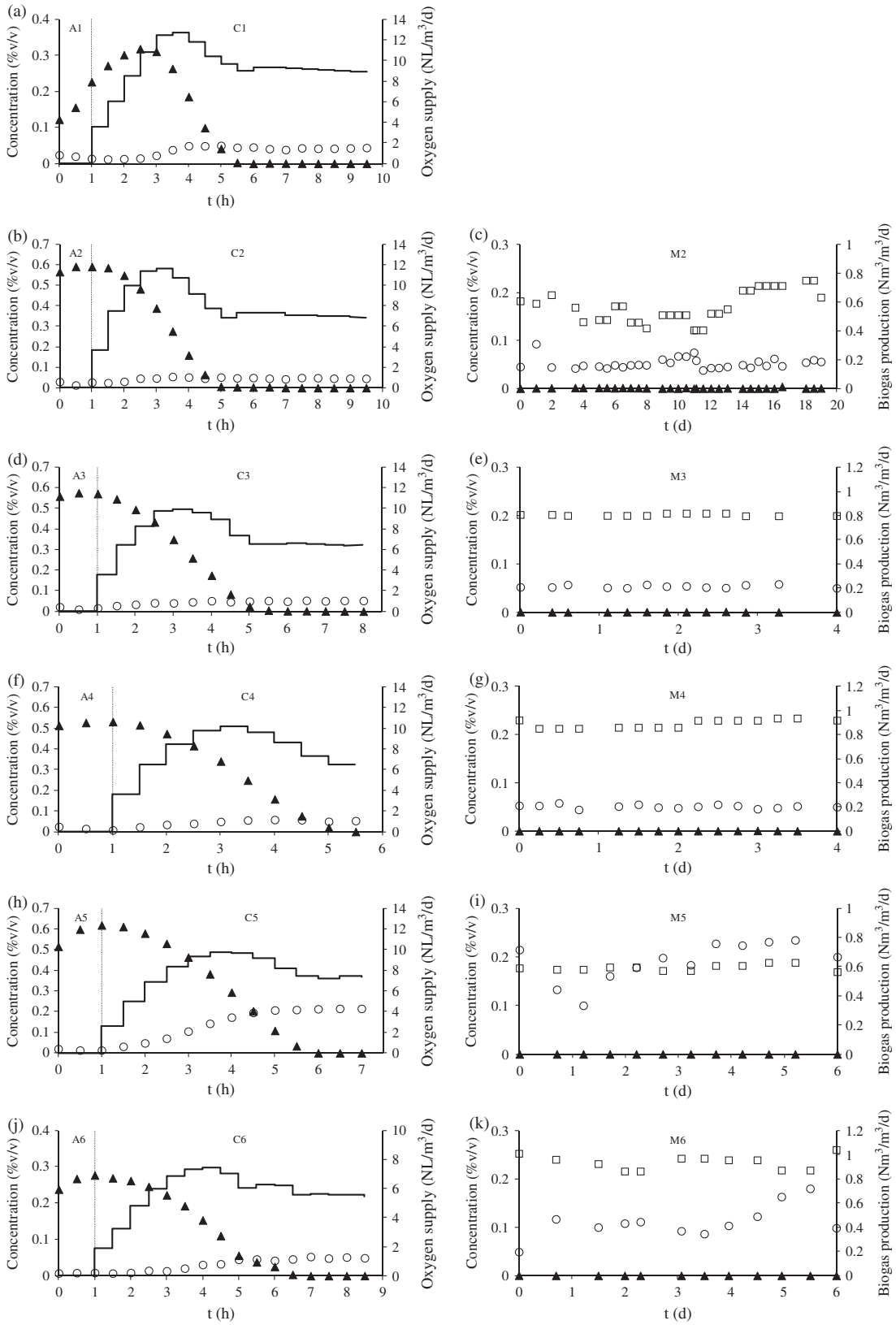


Fig. 2. Sulphide (▲) and oxygen (○) content of biogas, oxygen supply (continuous line) and biogas production (□) in trial 1 (a), 2 (b and c), 3 (d and e), 4 (f and g), 5 (h and i) and 6 (j and k) during S1.

Until the 17th day, the biogas produced in both digesters was entirely desulphurised (Fig. 3a and c). Moreover, although the oxygen flow rate was significantly reduced (by approximately 62%) due to the significant decrease in the feeding VS content and the resulting lower biogas production, the oxygen concentration remained fairly stable around 0.04 and 0.05%v/v in R1 and R2, respectively. This suggested that the hydrogen sulphide production certainly varied concurrently with both the biogas production and

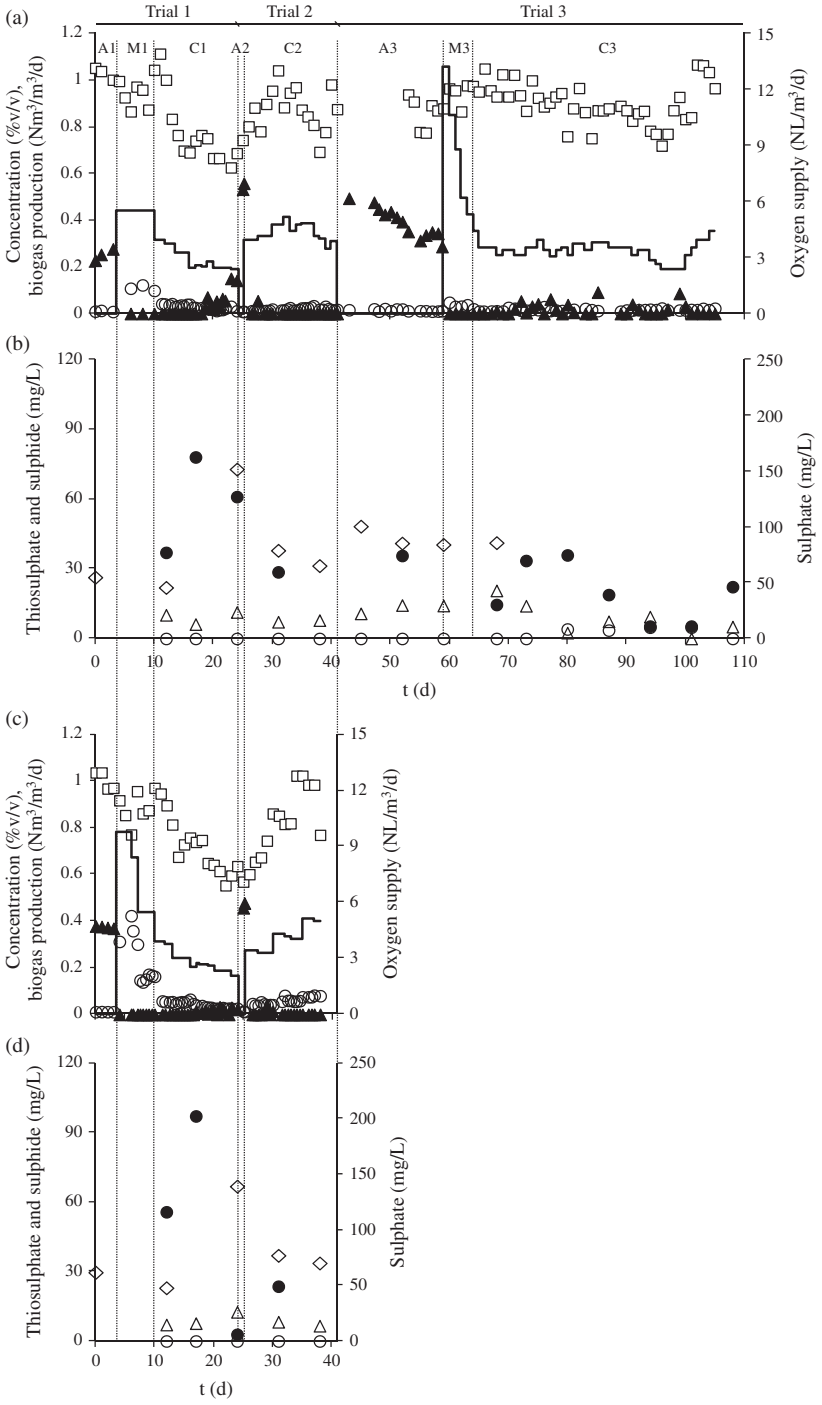


Fig. 3. Sulphide (▲) and oxygen (○) content of biogas, biogas production (□), oxygen flow rate (continuous line), sulphate (●) concentration in feed sludge, and sulphate (○), thiosulphate (▲) and sulphide (◇) concentration in R1 (a and b) and R2 (c and d) during S2.

the OLR (Table 2). Accordingly, and considering the above observations, the optimum ratio of oxygen supplied to gaseous sulphide produced was estimated at around 1.4 (v/v) for both digesters.

From the 17th day, some peaks of hydrogen sulphide concentration were detected, while the biogas oxygen content decreased appreciably. This pointed to a rise in the hydrogen sulphide concentration, which was confirmed in A2; it was around 0.54 and 0.47%v/v in R1 and R2, respectively (Table 2). This was at least partially attributed to the increase in the feeding sulphate content and the consequent rise in the sulphate-reducing activity, which was indeed consistent with the rise in the dissolved sulphide concentration observed on the 24th day (Fig. 3b and d). It should be noted that thiosulphate was rarely detected in the feed sludge, and its concentration did not exceed 10 mg/L; therefore, it was not depicted in Fig. 3.

After almost two days in the absence of oxygen (period A2 in Table 2), microaerobic conditions were restored. In C2, the oxygen supply to both digesters was regulated according to a relationship of 5.0 $NL_{oxygen}/N\ m^3_{biogas}$ in order to operate under an oxygen/hydrogen sulphide ratio similar to that maintained until the 17th day. This micro-oxygenation relationship proved to be the optimum, since the average removal efficiency was 100%, and minute amounts of

oxygen remained unused in the biogas (Fig. 3a and c). Furthermore, similarly to C1, the oxygen concentration remained fairly stable despite the substantial change in the oxygen supply. This was raised by approximately 65% in order to maintain the aforementioned micro-oxygenation relationship (Table 2). With regard to this, it is worth noting that the excess of oxygen was slightly higher in R2, which was related to the somewhat lower starting concentration of hydrogen sulphide in the biogas (see period A2).

Micro-oxygenation was interrupted for maintenance from the 41st to the 59th day (period A3 in Table 2). Once biogas sulphide content stabilised at approximately 0.33%v/v, the micro-oxygenation was restarted (period M3). It was supplied to the recirculation stream instead of to the gas space. Accordingly, a higher oxygen demand was expected in order to maintain performances similar to those achieved until then; hence the relatively larger initial dose (Fig. 3a). In fact, considering the oxygen concentrations in the biogas resulting from the significantly lower oxygen supplies maintained in the preceding microaerobic periods (where very similar starting values of both hydrogen sulphide concentration and biogas production were recorded), the oxygen consumption effectively increased. Nevertheless, since the biogas's oxygen content was slightly higher than in those periods, and the removal efficiency was 100%, the micro-oxygenation level was gradually lowered. A relationship of 3.5 $NL_{oxygen}/N\ m^3_{biogas}$ was eventually reached (65th day). This was used to regulate the oxygen flow rate during C3. Hence, the oxygen transfer to the liquid phase was minimised, thereby reaching rates similar to those maintained with oxygen supply to the headspace. Furthermore, as in the previous M and C periods, the methane concentration was not negatively affected.

At this point, it should be noted that Díaz et al. [22] found that the optimum micro-oxygenation relationship at 0.33%v/v of hydrogen sulphide concentration was 6.4 $NL_{oxygen}/N\ m^3_{biogas}$. The higher oxygen demand in comparison with this study was due to the fact that they operated with biogas recirculation instead of sludge recirculation. This resulted in an increased O_2 transfer to the liquid phase. In fact, in contrast to this research, dissolved sulphide was also removed, which was consistent with the previous findings [12].

In the last control period (C3 in Table 2), the mean removal efficiency was 95% (Fig. 3a); several peaks of hydrogen sulphide concentration were recorded. Some peaks, such as those detected around the 98th day, were ascribed to an insufficient regulation frequency of micro-oxygenation. The others probably resulted from a higher sulphate-reducing activity, since a substantial rise in the feeding sulphate content was recorded between the 73rd and the 80th day (Fig. 3b). In the meantime, the oxygen concentration in the biogas did not exceed 0.03%v/v, and remained even more stable than in the previous C periods (C1 and C2), which supported the assumption that the biogas and the hydrogen sulphide production both varied concurrently with the OLR. At this point, it should be highlighted that the oxygen supplies maintained in S2 were significantly tighter than in S1; hence the lower oxygen concentrations and larger amount of peaks of hydrogen sulphide concentration in the biogas produced during S2.

As shown in Fig. 3b and d, similarly low concentrations of thiosulphate were maintained all through S2. This suggested incomplete reduction of the sulphate contained in the feed sludge. Consequently, its presence inside both reactors in the M and C periods could not be attributed to the further oxidation of sulphide. On the other hand, sulphate was detected only inside R1 and in concentrations lower than 10 mg/L from the 80th day, which was linked to the change in the oxygen dosing point. Therefore, as in S1, elemental sulphur was presumably the main by-product of the hydrogen sulphide oxidation during S2. This was indeed consistent with the really limited micro-oxygenation levels maintained during the research.

3.3. Application proposals

The parameter for regulating the oxygen supply during microaerobic digestion in order to control the biogas sulphide content must be selected according to the operational conditions of the bioreactor. Under variable loading rates of both organic matter and sulphur (as in this study), or when the feedstock to the digester presents a variable composition, biogas sulphide content could be the basis for the development of a precise and consistent control strategy. A feedback PID control could be employed as the tool to continuously adjust the oxygen supply according to the hydrogen sulphide concentration. On the other hand, biogas production could be the basis for regulating the oxygen flow rate during digestion under variable OLR and steady sulphur load. Although biogas sulphide content could also be used under such conditions, it must be considered that monitoring the biogas flow rate is cheaper due to lower costs of the measuring equipment.

The optimum micro-oxygenation relationship (oxygen supply/biogas production) at a particular hydrogen sulphide concentration is specific for every plant, since it depends on the feedstock, the operational conditions, and even the reactor configuration. A readjustment of the micro-oxygenation is recommended at least daily. Clearly, the more often the oxygen supply is adjusted, the lower the probability of surpassing the concentration limit of hydrogen sulphide and of wasting oxygen, thereby ensuring an optimum biogas quality. Evidently, the readjustment periodicity would depend on the stringency required in the control. Similarly, the PID control could be run in discontinuous mode; after the oxygen flow rate was adjusted, the controller could be kept on standby, while the micro-oxygenation level could be maintained constant. Meantime, hydrogen sulphide concentration in biogas would be determined with the required periodicity. Thus, in the event of an increasing concentration or successive zeros, the controller would be restarted in order to re-optimise the oxygen supply.

4. Conclusions

This study regarded the suitability of using biogas production, and hydrogen sulphide concentration in biogas, in order to regulate the oxygen supply during microaerobic digestion for the development of precise and consistent control strategies was studied. Biogas production could be an efficient regulating parameter under variable OLR and steady sulphur load, independently of the oxygen dosing point. Under non-steady sulphur load, biogas sulphide content should be used instead. A feedback PID controller could be implemented in order to accurately adjust the oxygen flow rate according to the hydrogen sulphide concentration.

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Appendix A

A.1. Description of a feedback PID control

The basic structure of the feedback (or closed loop) PID controller in parallel form is shown in Fig. A.1. It assumes a linear relationship between the input and the output variable to the process (u and y) within a limited operational range involving small variations in both variables and around an equilibrium point

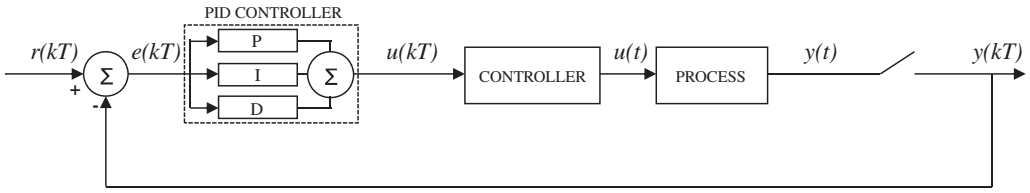


Fig. A.1. Feedback PID controller in parallel form.

Table A.1
Effects of independent P, I and D tuning (adapted from [24]).

	Drop time	Steady-state error	Stability
Increasing K_p	Decrease	Decrease	Degrade
Decreasing T_i	Small decrease	Large decrease	Degrade
Increasing T_d	Small decrease	Minor change	Improve

[26]. It acts (u) according to the control error (e), which is the difference between the process output or controlled variable (y) and the desired process output or set-point (r). The algorithm of this version of the PID controller is shown in Eq. (A.1), where K_p (proportional gain), T_i (integral time), and T_d (derivative time) are the control parameters [23]. It combines three types of actions; one of them (P in Fig. A.1) is proportional to $e(y - r)$, another (I) is proportional to the integral of e, and the last one (D) is proportional to the derivative of e.

$$u(t) = K_p\left(e(t) + \frac{1}{T_i}\int_0^t e(t)dt + T_d\frac{de(t)}{dt}\right) \quad (A.1)$$

In order to digitally implement this continuous-time control law, Eq. (A.1) must be discretised by defining a sampling time (T) [23]. So, it can be rewritten as:

$$u(t) = u(t-T) + g_0e(t) + g_1e(t-T) + g_2e(t-2T) \quad (A.2)$$

where g_0, g_1 and g_2, are the control constants, which depend on the control parameters according to Eqs. (A.3), (A.4) and (A.5). Therefore, the PID controller set the oxygen supply at every T based on the current control error, the accumulated control error, and the tendency of the control error.

$$g_0 = K_P\left(1 + \frac{T}{T_i} + \frac{T_d}{T}\right) \quad (A.3)$$

$$g_1 = K_P\left[-1 - \left(\frac{2T_d}{T}\right)\right] \quad (A.4)$$

$$g_2 = K_P\left(\frac{T_d}{T}\right) \quad (A.5)$$

The individual effects of the three functionalities of the PID control on three of the most important characteristics of the closed loop response of the controller are shown in Table A.1. "Drop time" reflects the time required to achieve the set-point, "steady-state error" is the residual error (namely, that which remains after the controlled variable has stabilised around the set-point), and the concept "stability" refers to operation within the linear range (see below). As suggested, the controller must keep the process variables within a specified range that ensures stability.

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